FLAT BASE CHANGE FOR THE LOWER SHRIEK OF THE COTANGENT SETTING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to two labelled equations of the chapter fixing the commutative and double squares; the proof takes the notation from the resolution remark.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The resolution remark describing the complexes and squares used. Remark remark-resolution, used at proof lines 1183-1186. Supplies the notation and the reductions taken for granted.
- The criterion computing the lower shriek from a resolution. Cohomology on Sites, Lemma sites-cohomology-lemma-get-it-now, used at proof lines 1185-1188. Reduces the claim to the quasi-isomorphism statement.
- The comparison between simplicial algebras and their associated complexes. used at proof lines 1188-1194. Identifies the derived tensor product with the complex of the base change.
- Freely generated resolutions of the algebra B. used at proof lines 1189-1192. Makes s(P_bullet) a free resolution, so the derived tensor product is computed as claimed.

Source reference. Original equation 08QM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11065–11065. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original equation 08QN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11066–11066. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 08QI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11061–11061. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08QA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11053–11053. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Assume (08QM) induces a quasi-isomorphism $B \otimes^{\mathbf{L}}_A A' = B'$. Then, with notation as in (08QN) and $\mathcal{F}' \in Ab(\mathcal{C}')$, we have $L\pi_!(g^{-1}\mathcal{F}') = L\pi'_!(\mathcal{F}')$.*

Proof. We use the results of Remark 08QI without further mention. We will apply Cohomology on Sites, Lemma 08QA. Let $P_\bullet \to B$ be a resolution. If we can show that $u(P_\bullet) = P_\bullet \otimes_A A' \to B'$ is a quasi-isomorphism, then we are done. The complex of A-modules $s(P_\bullet)$ associated to $P_\bullet$ (viewed as a simplicial A-module) is a free A-module resolution of B. Namely, P_n is a free A-module and $s(P_\bullet) \to B$ is a quasi-isomorphism. Thus $B \otimes^{\mathbf{L}}_A A'$ is computed by $s(P_\bullet) \otimes_A A' = s(P_\bullet \otimes_A A')$. Therefore the assumption of the lemma signifies that $\epsilon' : P_\bullet \otimes_A A' \to B'$ is a quasi-isomorphism. □